

Math 31B — Final Review: one map for the whole course | answers filled in

board layout, drawn live over 50 minutes | black = structure | gray = when you reach for it | red = the example you already did | dashed red = the three links, drawn as they come up

Say this at minute 0. Wednesday's exam prints the trig reduction formulas, the entire convergence-test table — ten rows, statements *and* hypotheses — and five Taylor series: e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $\arctan x$. $\sum_{n \geq 0} x^n = \frac{1}{1-x}$ is *not* one of those five; you read it off the geometric row of the test table instead. What the sheet never gives you is which tool to reach for — and that is what the next fifty minutes are about.

1 $\int f(x) dx$ — “what is it?”

u-substitution you can see g and g' sitting in the integrand

$$\text{ex } \int_1^2 \frac{x^2}{x^3+4} dx \text{ (ET2)} = \frac{1}{3} \ln \frac{12}{5}$$

Integration by parts two unrelated factors; or $\ln x$ / $\tan^{-1} x$ standing alone

$$\text{ex } \int_0^1 x e^{-2x} dx \text{ (ET3)} = \frac{1-3e^{-2}}{4}$$

Trig integrals powers of $\sin$, $\cos$, $\tan$, $\sec$ — reduction formulas are on the exam

$$\text{ex } \int \cos(7\theta) \cos^2(7\theta) d\theta \text{ (WS3 1e)} = \frac{1}{7} \sin 7\theta - \frac{1}{21} \sin^3 7\theta + C$$

Trig substitution $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$

$$\text{ex } \int_{-3/2}^{3/2} \sqrt{9-4x^2} dx \text{ (MT1 #9)} = \frac{9\pi}{4}$$

Partial fractions a rational function whose denominator factors

$$\text{ex } \int \frac{dx}{(x^2+10)(x-4)^2} \text{ (MT1 #2, form only)} = \frac{A}{x-4} + \frac{B}{(x-4)^2} + \frac{Cx+D}{x^2+10}$$

FTC part 1 + chain rule $\frac{d}{dx}$ of an integral with a variable limit

$$\text{ex } \frac{d}{dx} \int_0^{\ln x} \sqrt{1+t^4} dt \text{ (MT1 #6)} = \frac{\sqrt{1+(\ln x)^4}}{x}$$

Area / net change “area bounded by”, “net change” — split where the curves cross

$$\text{ex } \text{area between } x^3 \text{ and } x \text{ on } [-1, 1] \text{ (MT1 #10)} = \frac{1}{2}$$

2 Improper $\int$ — “does it exist?”

Rewrite as a limit — always line one ①

$\int_1^\infty = \lim_{b \rightarrow \infty} \int_1^b$, and split at every bad point

ex $\int_{-\infty}^0 e^x dx$ (ET5) = 1; the limit notation *is* the five points

p-test at ∞ : converges iff $p > 1$; at 0: iff $p < 1$.

Opposite directions — check which end is bad.

ex $\int_0^1 \frac{dx}{x^5}$ (MT2 1c) diverges

Comparison bound *above* to prove convergence, *below* to prove divergence

ex $\int_1^\infty \frac{dx}{x^2 + \sin^2 x}$ (WS5 #5) $\leq \int_1^\infty \frac{dx}{x^2}$: converges

WS5 #7 (Herbert) bounds above and then **wrongly** concludes divergence: an upper bound can never prove divergence. The direction is the entire problem.

3 $\{a_n\}$ — “does $a_n \rightarrow L$?”

Pass to $f(x)$, then L'Hôpital a sequence has no derivative: move to f with $a_n = f(n)$, take the limit, come back. One-way: f having no limit settles nothing about a_n .
ex $\frac{\ln n}{n}$ (ET6) $\rightarrow 0$

Squeeze a $\sin$, $\cos$ or $n!$ you cannot evaluate, trapped between two limits you can

$$\text{ex } 1 \leq a_n \leq 2^{1/n} \text{ (MT2 #9)} \rightarrow 1$$

$$\text{ex } \frac{\cos n}{n!} \text{ (WS6 #6)} \rightarrow 0$$

The sign: $|a_n|$ is not a_n ② $|a_n| \rightarrow L$ does *not* give $a_n \rightarrow L$ — *except* when $L = 0$

ex $\cos(\pi n)$ (WS6 2d): $= (-1)^n$, so $|a_n| \rightarrow 1$ but the limit does not exist

Geometric ratio $r^n \rightarrow 0$ exactly when $|r| < 1$

$$\text{ex } \frac{3^{n+2}}{5^n} \text{ (MT2 8b)} \rightarrow 0$$

4 $\sum a_n$ — “does S_N converge?” (and only sometimes: “what is the sum?”)

A. You can produce the sum.

Geometric cr^n with $c \neq 0$: converges exactly when $|r| < 1$, to $\frac{\text{first term}}{1-r}$ — read the first term off the term you actually have, not off $n = 0$

$$\text{ex } \sum_{n \geq 1} (2/3)^n \text{ (WS7 1a)} = 2; \quad \sum_{n \geq 0} \frac{3^{n+1} - 2(-1)^n}{2^{2n}} \text{ (MT2 #10)} = \frac{52}{5}$$

Telescoping / partial sums consecutive terms cancel — write S_N and let $N \rightarrow \infty$

$$\text{ex } \sum_{n \geq 7} (\frac{1}{n} - \frac{1}{n+1}) \text{ (WS7 1c)} = \frac{1}{7}; \quad \sum_{n \geq 1} (\sqrt{n+1} - \sqrt{n}) \text{ (ET7): } S_N = \sqrt{N+1} - 1 \rightarrow \infty, \text{ diverges } \textit{although } a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0$$

B. You can only answer yes or no.

Divergence Test — run it first, every time ② if $a_n \not\rightarrow 0$ it diverges; if $a_n \rightarrow 0$ you have learned nothing and must keep going

$$\text{ex } \sum (-1)^{n+1} \frac{n}{n+1} \text{ (WS8 #2): } a_n \text{ has no limit, so the series diverges}$$

p-series, and the Integral Test ① for the Integral Test say *positive, continuous, decreasing* out loud — a graded step

$$\text{ex } \sum_{n \geq 2} \frac{1}{n(\ln n)^2} \text{ (WS7 2b): converges, since } \int_2^\infty \frac{dx}{x(\ln x)^2} = \frac{1}{\ln 2} \text{ is finite}$$

Direct and Limit Comparison the terms *look like* a p -series or a geometric once you keep only the largest piece. Both need positive terms; for LCT also say $\lim a_n/b_n$ is finite and positive.

$$\text{ex } \sum \frac{2+\sin n}{n\sqrt{n+\ln n}} \text{ (MT2 12a): } \leq \sum \frac{3}{n^{3/2}}, \text{ converges; } \sum \frac{n}{n^2+1} \text{ (WS7 2e): against } \frac{1}{n}, \text{ diverges}$$

Alternating Series Test — but step one is always $\sum |a_n|$ if it converges you are done and the answer is *absolute*; only when it fails do you reach for the AST

$$\text{ex } \sum_{n \geq 1} \frac{(-1)^{n+8}}{n^2+4} \text{ (ET8): } \frac{8}{n^2+4} < \frac{8}{n^2}, \text{ so } \textit{absolute}; \quad \sum_{n \geq 2} \frac{(-1)^n}{n \ln n} \text{ (WS8 3a): step one fails, AST works, so } \textit{conditional}$$

Ratio and Root factorials, n th powers, c^n . The statement is printed; *when* to reach for it is not. $L = 1$ says nothing; $L = 0$ is a fine $L < 1$.

$$\text{ex } \sum \frac{3^n}{(n+1)!} \text{ (ET9): ratio} = \frac{3}{n+2} \rightarrow 0, \text{ converges; } \sum \frac{1}{n} \text{ and } \sum \frac{1}{n^2} \text{ (WS9 #1): the same } L = 1, \text{ opposite verdicts}$$

5 $\sum a_n(x-c)^n$ — “for which x ?”

Interval of convergence Ratio Test on the *terms* gives R . R is not the answer — then test each endpoint by hand. Putting $x = c \pm R$ into the whole term $a_n(x-c)^n$ turns the power series into an ordinary number series; test *that*, never $\sum a_n$ on its own.

ex $\sum_{n \geq 1} \frac{3^n(x-2)^n}{n4^n}$ (WS9 5a): $R = \frac{4}{3}$, interval $[\frac{2}{3}, \frac{10}{3})$ — at the left end $\sum_{n \geq 1} \frac{(-1)^n}{n}$ converges, at the right end $\sum_{n \geq 1} \frac{1}{n}$ diverges

ex $\sum_{n \geq 1} \frac{n!}{5^n} (x-3)^n$ (WS9 5b): $R = 0$, so it converges only at $x = 3$ and there are no endpoints to test

Build it from what you are given ③ five Taylor series are printed. $\frac{1}{1-x} = \sum_{n \geq 0} x^n$ for $|x| < 1$ is not among them — it is the geometric row of the test table, and it is what WS9 #4, WS10 #2 and ET10 all run on. Four moves: substitute, multiply, differentiate, integrate.

ex $\frac{1}{x}$ at $c = 1$ (ET10): write $x = 1 + (x-1)$; the denominator is then $1-r$ with $r = -(x-1)$, so $\frac{1}{x} = \sum_{n \geq 0} (-1)^n (x-1)^n$ on $|x-1| < 1$

ex $(x^2+2x)e^{3x}$ at $c = 0$ (Lec 13 HW 5d): substitute $3x$ into e^x , then multiply: $\sum_{n \geq 0} \frac{3^n}{n!} (x^{n+2} + 2x^{n+1})$, all x

Coefficient $\leftrightarrow$ derivative $a_n = \frac{f^{(n)}(c)}{n!}$, so $f^{(n)}(c) = n! a_n$. A coefficient is a number, not a derivative.

$$\text{ex } \text{from } \sum \frac{(n+1)x^n}{3^n} \text{ (Lec 13 HW 4): } f^{(99)}(0) = 99! \cdot \frac{100}{3^{99}}$$

① **2 $\leftrightarrow$ 4** the one place an improper integral answers a series question: $\sum_{n \geq 2} \frac{1}{n(\ln n)^2}$ becomes $\int_2^\infty \frac{dx}{x(\ln x)^2}$. ② **3 $\leftrightarrow$ 4** $\sum a_n$ converges *means* $\{S_N\}$ converges, so station 3's tools are available for S_N ; and the Divergence Test is a station-3 limit settling a station-4 question. ③ **1 $\leftrightarrow$ 5** differentiate or integrate a power series termwise on $|x| < R$: $\frac{1}{1-x^4} \mapsto \frac{4x^3}{(1-x^4)^2}$ and $\frac{1}{1+2x} \mapsto \frac{\ln(1+2x)}{2}$ (Lec 13 HW 1, 2). The radius holds, but an endpoint can move — the second gains $x = \frac{1}{2}$.

Building the map — the fifty minutes

draw order, what to say, and what to work in full. The map is built left to right, top row then bottom row; nothing is erased.

Min	What goes on the board	The sentence that carries it	Worked in full
0–3	The five headers and the five questions. Nothing underneath them yet.	“Every problem on Wednesday is one of these five questions. Which question you were asked narrows you to one station; what the problem then hands you picks the technique inside it. That two-step choice is what is being tested.”	—
3–12	Station 1, seven branches, top to bottom. Write the gray trigger <i>before</i> the technique name.	The trigger is what you are handed; the technique is what you are trying to recover from it.	MT1 #9, trig substitution. The one station-1 example worth five minutes.
12–19	Station 2, three branches.	Write the limit line even when the integral is easy — ET5 is five points of notation and nothing else.	ET5; write $\lim_{a \rightarrow -\infty}$ <i>above</i> it in a second color, so the notation is visibly the answer.
19–26	Station 3, four branches.	A sequence has no derivative. That is the whole reason you move to $f(x)$ and the whole reason you must move back.	ET6.
26–41	Station 4. Tier A (two branches), then Tier B (five). Draw link ② as you write the Divergence Test (≈ 29), and ① as you write the Integral Test (≈ 32).	Write “Step 1” on the board as a literal numbered step. Step 1 is $\sum a_n $, every time, before the word <i>alternating</i> is allowed to appear.	ET8 in full, then WS8 3(a) written <i>directly beside it</i> , so absolute and conditional stand side by side.
41–49	Station 5, three branches. Draw link ③ when you reach “build it from what you are given” (≈ 45).	R is not the answer; the interval is. When R is finite and positive there are two endpoints, and they are two separate ordinary number series.	WS9 5(a), including both endpoints written out.
49–50	Stop writing. Step back and point at the three dashed links.	The three sentences printed at the foot of page 1.	—

Checkpoints. Station 2 at 12, station 4 at 26, station 5 at 41. More than two minutes late: cut, do not speed up — speeding up costs the copying, and the copy is the deliverable.

Cut order. (1) Station 1’s *trig integrals* and *area / net change* branches, combined into a single line with no worked example. (2) The second example under station 4’s geometric branch (MT2 #10), and the second under station 3’s Squeeze branch (WS6 #6); MT2 #9 alone still carries Squeeze. (3) The WS9 #1 pair under Ratio and Root: state that $L = 1$ tells you nothing without working both. **Do not** cut station 3’s *geometric ratio* branch: $r^n \rightarrow 0$ is about a sequence, $\sum cr^n$ about a series, and MT2 8b needs the first.

Never cut station 4’s Step-1 example or station 5’s endpoint example — the two places students lose points while writing correct mathematics.

The five examples you work in full — board-ready, so nothing is re-derived live

MT1 #9 (min 3–12)

$\int_{-3/2}^{3/2} \sqrt{9-4x^2} dx$. Put $2x = 3 \sin \theta$:
 $dx = \frac{3}{2} \cos \theta d\theta$, $\sqrt{9-4x^2} = 3 \cos \theta$, and
 $x = \pm \frac{3}{2} \Rightarrow \theta = \pm \frac{\pi}{2}$. The integral becomes
 $\frac{9}{2} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta$, and $\int \cos^2$ is printed on the exam, so the value is $\frac{9}{2} \cdot \frac{\pi}{2} = \frac{9\pi}{4}$.
 Say: the substitution is chosen by the *shape under the root*, and the limits move with it.

ET5 (min 12–19)

$\int_{-\infty}^0 e^x dx =$
 $\lim_{a \rightarrow -\infty} \int_a^0 e^x dx =$
 $\lim_{a \rightarrow -\infty} (e^0 - e^a) = 1 - 0 = 1$.
 Say: the algebra is one line. The limit symbol is what is being graded, so it goes down before anything else.

ET6 (min 19–26)

$a_n = \frac{\ln n}{n}$ has no derivative, so set $f(x) = \frac{\ln x}{x}$ with $a_n = f(n)$. Then $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$ is $\frac{\infty}{\infty}$, and L’Hôpital gives $\lim \frac{1/x}{1} = 0$. Hand it back: $a_n \rightarrow 0$, so the sequence converges to 0.
 Say the last step out loud — the limit was computed for f , and it has to be handed back to the sequence.

ET8, with WS8 3(a) beside it (min 26–41)

Left. $\sum \frac{(-1)^n 8}{n^2+4}$. **Step 1:** for $n \geq 1$, $0 < \frac{8}{n^2+4} < \frac{8}{n^2}$ and $\sum \frac{8}{n^2}$ converges ($p = 2$), so $\sum |a_n|$ converges by direct comparison. **Absolutely convergent** — the AST is never needed.
Right. $\sum_{n \geq 2} \frac{(-1)^n}{n \ln n}$. **Step 1:** $f(x) = \frac{1}{x \ln x}$ is positive, continuous and decreasing on $[2, \infty)$, and $\int_2^\infty \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} \int_2^b \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} [\ln \ln x]_2^b = \infty$, so by the **Integral Test** $\sum |a_n|$ diverges and step 1 *fails*. **Step 2:** $b_n = \frac{1}{n \ln n}$ is positive, decreasing and $\rightarrow 0$, so the AST applies. **Conditionally convergent.**

Keep both on the board at once: the verdicts differ only in whether step 1 succeeded.

WS9 5(a) (min 41–49)

$\sum_{n \geq 1} t_n$ with $t_n = \frac{3^n(x-2)^n}{n4^n}$. Ratio of consecutive terms:
 $\left| \frac{t_{n+1}}{t_n} \right| = \frac{3|x-2|}{4} \cdot \frac{n}{n+1} \rightarrow \frac{3}{4}|x-2|$, which is < 1 exactly when $|x-2| < \frac{4}{3}$. So $R = \frac{4}{3}$ about the center 2. At $x = \frac{2}{3}$ the term is $\frac{3^n(-4/3)^n}{n4^n} = \frac{(-1)^n}{n}$, and $b_n = \frac{1}{n}$ is positive, decreasing and $\rightarrow 0$, so the AST gives convergence: **include**. At $x = \frac{10}{3}$ the term is $\frac{1}{n}$; diverges ($p = 1$), so **exclude**. Interval $[\frac{2}{3}, \frac{10}{3})$.
 Say: substituting an endpoint turns the power series back into an ordinary number series, and every tool in station 4 is available again.

Tell them this at minute 0, before you draw anything

The whole map fits on one page and they should copy it. The red items are all problems they have already done, so they write only the label — ET8, WS9 5(a), MT1 #9 — and look the problem up tonight. Otherwise the hour goes into transcribing integrals instead of watching the structure being built.

Three colors, no more

Black for the structure, one color for the trigger, one for the example. With only two markers, drop the trigger color and keep the example color.

Three questions to expect, with answers ready

“Should I just use the Ratio Test on everything?” No, and the counterexample is already on their sheet: WS9 #1 gives $L = 1$ on both $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$. An $L = 1$ costs the whole attempt and you start over.

“Do we have to memorize the Taylor series?” Five are printed. $\frac{1}{1-x}$ is not one of the five — you read it off the geometric row of the test table, or you just know it.

“Does the Integral Test give the sum?” No. $\sum_{n \geq 2} \frac{1}{n(\ln n)^2}$ is not $\frac{1}{\ln 2}$; the integral only had to come out finite. A test on this list answers yes or no. In this course the sum itself comes from one of three places: a geometric series, a telescoping partial sum, or a series you recognize as one of the Taylor series.